

Artemy Kolchinsky

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Last updated: June 28, 2026

RESEARCH PROFILE I am a theorist working at the interface of nonequilibrium statistical physics and information theory. My work focuses on fundamental bounds, tradeoffs, and decompositions that govern information processing in nonequilibrium systems. I apply these ideas to understand adaptation and agency in systems ranging from molecular replicators to neural networks.

RESEARCH POSITIONS

2023–	Universitat Pompeu Fabra (Barcelona, Spain). Marie Curie Fellow.	Host: Ricard Solé
2022–2023	University of Tokyo, Universal Biology Institute. Project Researcher.	PI: Sosuke Ito
2015–2021	Santa Fe Institute (Santa Fe, USA). Program Postdoctoral Fellow.	PI: David H. Wolpert
2014	LinkedIn Corporation, Data Science Research Intern.	
2009–2010	Instituto Gulbenkian de Ciência (Oeiras, Portugal). Predoctoral Visiting Researcher.	

AFFILIATIONS

2024 –	Barcelona Collaboratorium for Modelling and Predictive Biology. Associated fellow.
2023 –	University of Tokyo, Universal Biology Institute. Affiliated researcher.

EDUCATION

2015	Indiana University. Ph.D. in Informatics (Complex Systems focus), Minor in Cognitive Science Thesis: “Measuring scales: integration and modularity in complex systems”. Advisor: Luis M. Rocha
2004	New York University, B.A. <i>Magna Cum Laude</i>

PUBLICATIONS *Equal contribution †Corresponding author

A. Kolchinsky, “Elementary derivation of the dissipation–coherence bound for stochastic oscillators”, *Physical Review E*, in press. [arXiv:2510.14101](https://arxiv.org/abs/2510.14101) [link](#)

J. Bauermann, G. Bartolucci, **A. Kolchinsky**, “Spatiotemporal organization of chemical oscillators via phase separation”, *Physical Review Letters*, 136, 178001 (2026). [pdf link](#)

A. Kolchinsky[†], A. Dechant, K. Yoshimura, S. Ito, “Generalized free energy and excess/housekeeping decomposition in nonequilibrium systems: from large deviations to thermodynamic speed limits”, *Physical Review Research*, 8, 023025 (2026). [pdf link](#)

J. Piñero, D. R. Sowinski, G. Ghoshal, A. Frank, **A. Kolchinsky**[†], “Information bounds production in replicator systems”, *Communications Physics*, 9, 120 (2026). [pdf link](#)

M. Aguilera^{*†}, S. Ito, **A. Kolchinsky**^{*†}, “Inferring entropy production in many-body systems using nonequilibrium maximum entropy”, *Physical Review Letters*, 136, 077101 (2026). [pdf link](#)

OoLEN collaboration, “What it takes to solve the Origin(s) of Life: An integrated review. Part 1: Experimental methods and data repositories”, *Cell Reports Physical Science* (2026). [pdf link](#)

OoLEN collaboration, “What it takes to solve the Origin(s) of Life: An integrated review. Part 2: Theoretical methods and emerging trends”, *Cell Reports Physical Science* (2026). [pdf link](#)

G.-H. Xu, **A. Kolchinsky**, J.-C. Delvenne, S. Ito, “Thermodynamic geometric constraint on the spectrum of Markov rate matrices”, *Physical Review Letters*, 135, 257102 (2025). [pdf link](#) (*Editors’ Suggestion*)

J. Piñero^{*}, **A. Kolchinsky**^{*}, S. Redner, R. Solé, “Neutral theory of cooperative dynamics”, *Proceedings of the National Academy of Sciences*, 122, 51 (2025). [pdf link](#)

A. Kolchinsky, “Thermodynamics of Darwinian evolution in molecular replicators”, *Philosophical Transactions of the Royal Society B* 380, 20240436 (2025). [pdf link](#)

K. R. Thomsen, **A. Kolchinsky**, S. Rasmussen, “Protocellular energy transduction, information, and fitness”, *Philosophical Transactions of the Royal Society B* 380, 20240294 (2025). [pdf link](#)

S. Bartlett, A. W. Eckford, M. Egbert, M. Lingam, **A. Kolchinsky**[†], A. Frank[†], G. Ghoshal[†], “The Physics of Life: Exploring Information as a Distinctive Feature of Living Systems”, *PRX Life* 3, 037003 (2025). [pdf link](#)

R. Nagayama, K. Yoshimura, **A. Kolchinsky**, S. Ito, “Geometric thermodynamics of reaction-diffusion systems: Thermodynamic trade-off relations and optimal transport for pattern formation”, *Physical Review Research* 7, 033011 (2025). [pdf link](#)

A. Kolchinsky[†], I. Marvian, C. Gokler, Z. Liu, P. Shor, O. Shtanko, K. Thompson, D. Wolpert, S. Lloyd, “Maximizing free energy gain”, *Entropy* 27, 91 (2025). [pdf link](#)

A. Kolchinsky, “Thermodynamic dissipation does not bound replicator growth and decay rates”, *The Journal of Chemical Physics* 161, 124101 (2024). [pdf link](#) (2024 JCP Emerging Investigators Special Collection)

R. Solé, C. P. Kempes, B. Corominas-Murtra, M. De Domenico, **A. Kolchinsky**, M. Lachmann, E. Libby, S. Saavedra, E., D. Wolpert, “Fundamental constraints to the logic of living systems”, *Royal Society Interface Focus* 14, 20240010 (2024). [pdf link](#)

P.M. Riechers, C. Gupta, **A. Kolchinsky**, M. Gu, “Thermodynamically ideal quantum state inputs to any device”, *PRX Quantum* 5, 030318 (2024). [pdf link](#)

A. Kolchinsky, “Partial information decomposition: redundancy as information bottleneck”, *Entropy* 26(7), 546 (2024). [pdf link](#)

J. Piñero, R. Solé, **A. Kolchinsky**[†], “Optimization of nonequilibrium free energy harvesting illustrated on bacteriorhodopsin”, *Physical Review Research* 6, 013275 (2024). [pdf link](#)

A. Kolchinsky[†], N. Ohga, S. Ito, “Thermodynamic bound on spectral perturbations, with applications to oscillations and relaxation dynamics”, *Physical Review Research* 6, 013082 (2024). [pdf link](#)

D.R. Sowinski, J. Carroll-Nellenback, R.N. Markwick, J. Piñero, M. Gleiser, **A. Kolchinsky**, G. Ghoshal, A. Frank, “Semantic Information in a model of resource gathering agents”, *PRX Life* 1, 023003 (2023). [pdf link](#)

A. Kolchinsky[†], “Generalized Zurek’s bound on the cost of an individual classical or quantum computation”, *Physical Review E* 108, 034101 (2023). [pdf link](#)

M. Aguilera and **A. Kolchinsky**, “Quantifying higher-order entropy production in organized nonequilibrium states”, *Proceedings of the ALIFE 2023*, 45 (2023). [pdf link](#)

N. Ohga, S. Ito, **A. Kolchinsky**, “Thermodynamic bound on the asymmetry of cross-correlations”, *Physical Review Letters* 131, 077101 (2023). [pdf link](#) (Editors’ Suggestion; Featured in *Physics*)

K. Yoshimura, **A. Kolchinsky**, A. Dechant, S. Ito, “Housekeeping and excess entropy production for general nonlinear dynamics”, *Physical Review Research* 5, 013017 (2023). [pdf link](#)

F.C. Sheldon, **A. Kolchinsky**, F. Caravelli, “Computational capacity of LRC, memristive, and hybrid reservoirs”, *Physical Review E* 106, 045310 (2022). [pdf link](#)

A. Kolchinsky, “A novel approach to the partial information decomposition”, *Entropy* 24(3), 403 (2022). [pdf code link](#)

A. Kolchinsky[†] and D.H. Wolpert, “Dependence of integrated, instantaneous, and fluctuating entropy production on the initial state in quantum and classical processes”, *Physical Review E* 104, 054107 (2021). [pdf link](#)

A. Kolchinsky[†] and D.H. Wolpert, “Work, entropy production, and thermodynamics of information under protocol constraints”, *Physical Review X* 11, 041024 (2021). [pdf link](#)

A. Kolchinsky[†] and D.H. Wolpert, “Entropy production given constraints on the energy functions”, *Physical Review E* 104, 034129 (2021). [pdf link](#)

A. Kolchinsky and D.H. Wolpert, “Thermodynamic costs of Turing machines”, *Physical Review Research* 2, 033312 (2020). [pdf link](#)

D.H. Wolpert and **A. Kolchinsky**, “The thermodynamics of computing with circuits”, *New Journal of Physics* 22, 063047 (2020). [pdf link](#)

A. Kolchinsky and B. Corominas-Murtra, “Decomposing information into copying versus transformation”, *Royal Society Interface* 17, 20190623 (2020). [pdf link](#)

A.M. Saxe, Y. Bansal, J. Dapello, M. Advani, **A. Kolchinsky**, B.D. Tracey, D.D. Cox, “On the information bottleneck theory of deep learning”, *Journal of Statistical Mechanics*, 124020 (2019). [pdf code link](#)

- A. Kolchinsky**[†], B.D. Tracey, D.H. Wolpert, “Nonlinear information bottleneck”, *Entropy* 21(12), 1181 (2019). (*Entropy* 2021 Best Paper Award) [pdf code link](#)
- A. Berdahl, C. Brelsford, C. De Bacco, M. Dumas, V. Ferdinand, J.A. Grochow, L. Hébert-Dufresne, Y. Kallus, C.P. Kempes, **A. Kolchinsky**, D. B. Larremore, E. Libby, E.A. Power, C.A. Stern, B.D. Tracey, “Dynamics of beneficial epidemics”, *Scientific Reports* 9, 15093 (2019). [pdf link](#)
- E.A. Hobson, V. Ferdinand, **A. Kolchinsky**, J. Garland, “Rethinking animal social complexity measures with the help of complex systems concepts”, *Animal Behaviour* 155, 287-296 (2019). [pdf link](#)
- A. Kolchinsky**[†], B.D. Tracey, S. Van Kuyk, “Caveats for information bottleneck in deterministic scenarios”, *ICLR* (2019). [pdf code link](#)
- D.H. Wolpert, **A. Kolchinsky**, J.A. Owen, “A space–time tradeoff for implementing a function with master equation dynamics”, *Nature Communications* 10, 1727 (2019). [pdf link](#)
- A. Avena-Koenigsberger, X. Yan, **A. Kolchinsky**, M. van den Heuvel, P. Hagmann, O. Sporns, “A spectrum of routing strategies for brain networks”, *PLoS Computational Biology* 15(3): e1006833 (2019). [pdf link](#)
- J.A. Owen, **A. Kolchinsky**, D.H. Wolpert, “Number of hidden states needed to physically implement a given conditional distribution”, *New Journal of Physics* 21, 013022 (2019). [pdf link](#) ([correction](#))
- A. Kolchinsky**[†] and D.H. Wolpert, “Semantic information, autonomous agency, and nonequilibrium statistical physics”, *Royal Society Interface Focus* 8, 20180041 (2018). [pdf code link](#)
- A.M. Saxe, Y. Bansal, J. Dapello, M. Advani, **A. Kolchinsky**, B.D. Tracey, D.D. Cox, “On the information bottleneck theory of deep learning”, *ICLR* (2018). [pdf code link](#) (conference version)
- A. Kolchinsky**, N. Dhande, K. Park, Y.Y. Ahn, “The Minor Fall, the Major Lift: Inferring emotional valence of musical chords through lyrics”, *Royal Society Open Science* 4, 170952 (2017). [pdf data code link](#)
- A. Kolchinsky**[†], D.H. Wolpert, “Dependence of dissipation on the initial distribution over states”, *Journal of Statistical Mechanics*, 083202 (2017). [pdf link](#)
- A. Kolchinsky**, B.D. Tracey, “Estimating mixture entropy with pairwise distances”, *Entropy* 19(7), 361 (2017). ([correction](#)) [pdf code link](#)
- A. Kolchinsky**[†], A.J. Gates, L.M. Rocha, “Modularity and the spread of perturbations in complex dynamical systems,” *Physical Review E* 92, 060801(R) (2015). [pdf code link](#)
- A. Kolchinsky**, A. Lourenço, H. Wu, L. Li, L.M. Rocha, “Extraction of pharmacokinetic evidence of drug-drug interactions from the literature,” *PLOS One* 10(5): e0122199 (2015). [pdf link](#)
- A. Kolchinsky**, M.P. van den Heuvel, A. Griffa, P. Hagmann, L.M. Rocha, O. Sporns, J. Goñi, “Multi-scale integration and predictability in resting state brain activity,” *Frontiers in Neuroinformatics* 8, 66 (2014). [pdf link](#)
- A. Rossi, F.J. Parada, **A. Kolchinsky**, A. Puce, “Neural correlates of apparent motion perception of impoverished facial stimuli I: A comparison of ERP and ERSP activity,” *NeuroImage* 98, 442-459 (2014). [pdf link](#)
- A. Kolchinsky**, A. Lourenço, L. Li, L.M. Rocha, “Evaluation of linear classifiers on articles containing pharmacokinetic evidence of drug-drug interactions,” *Proc Pacific Symposium on Biocomputing*, 409-420 (2013). [pdf link](#)
- A. Kolchinsky** and L.M. Rocha, “Prediction and modularity in dynamical systems,” *Proc of European Conf. on the Synthesis and Simulation of Living Systems (ECAL)*, 65 (2011). [pdf link](#)
- A. Kolchinsky**, A. Abi-Haidar, J. Kaur, A.A. Hamed, L.M. Rocha, “Classification of protein-protein interaction full-text documents using text and citation network features,” *IEEE/ACM Transactions on Computational Biology and Bioinformatics* 7(3), 400-411 (2010). [pdf link](#)

PREPRINTS

- R. Nagayama, **A. Kolchinsky**, S. Ito, “Oscillatory-nonnormal decomposition of dissipation in Ornstein-Uhlenbeck processes”, [arXiv:2606.07263](#) (2026).
- O. M. Miangolarra, R. Sabbagh, **A. Kolchinsky**, “Wasserstein-2 gradient flows and the geometry of entropy production in classical and quantum stochastic thermodynamics”, [arXiv:2606.00698](#) (2026).
- A. Kolchinsky**, N. Ohga, S. Ito, “Cycle affinity and winding localize eigenvalues of Markov generators”, [arXiv:2605.15884](#) (2026).

J. Korbel, **A. Kolchinsky**, S.A.M. Loos, G. Manzano, R. Garcia-Millan, O. Movilla Miangolarra, É. Roldán, “Quo vadis, stochastic thermodynamics?”, [arXiv:2604.26601](https://arxiv.org/abs/2604.26601) (2026).

AWARDS & FUNDING

Fellowships and awards

- 2026 Ramón y Cajal Fellowship (RYC 2025), provisionally selected. Spanish State Research Agency (AEI)
- 2022 Marie Skłodowska-Curie Individual Fellowship (HORIZON-MSCA-2021-PF-01, #101068029) “NETOLIFE: Nonequilibrium thermodynamics of the origin of life”
- 2008– Lilly Fellowship in Informatics (competitive doctoral fellowship)
- 2009 Indiana University, funded by the Eli Lilly & Company Foundation

Research funding

- 2023 John Templeton Foundation (#62828), \$947,926, Co-PI “Goal-directed behavior and the origin of life”
- 2022 John Templeton Foundation (#62417), \$627,227, Co-PI “Information architectures that enable life: the emergence of meaning”
- 2019 Foundational Questions Institute (FQXi-RFP-IPW-1912), \$118,100, Co-investigator (PI: David H. Wolpert) “The role of constraints in the thermodynamics of intelligence”
- 2016 Foundational Questions Institute (FQXi-RFP-1622), \$128,319, Co-investigator (PI: David H. Wolpert) “Observers as self-maintaining non-equilibrium systems”
- 2016 NSF INSPIRE (CHE-1648973), \$999,947, Co-investigator (PI: David H. Wolpert) “Tradeoffs in the thermodynamics of computation: a new paradigm for biological information-processing”
- 2016 NSF (1620462), \$770,000, Co-investigator (PI: David H. Wolpert) “Information Networks and the Evolution of Social Organizations”

INVITED TALKS

- 5/2026 *Workshop on Stochastic Thermodynamics VII* (online)
- 3/2026 *Seminar series*, Instituto de Física Interdisciplinar y Sistemas Complejos (IFISC), Mallorca, Spain
- 11/2025 *Kyoto Workshop on Quantum Thermodynamics & Stochastic Thermodynamics*, Kyoto University, Japan
- 10/2025 *Autocatalysis in Reaction Networks Seminar (ARN)* (online)
- 10/2025 *Renormalization for AI Safety Workshop*, Boston, MA, USA
- 9/2025 *ILIAD2: ODYSSEY, conference on mathematical alignment*, Berkeley, CA, USA
- 6/2025 *CRC Colloquium*, Free University of Berlin, Berlin, Germany
- 5/2025 *Joint European Thermodynamics Conference*, Belgrade, Serbia
- 11/2024 *Universal Biology Institute Seminar Series*, University of Tokyo, Japan
- 9/2024 *Information in Matter Colloquium*, AMOLF, Amsterdam, Netherlands
- 7/2024 *Physics Department Seminar*, University of Barcelona, Barcelona, Spain
- 2/2024 *Seminar*, Barcelona Collaboratorium for Modelling and Predictive Biology, Barcelona, Spain
- 11/2023 *Mini-workshop: Non-equilibrium thermodynamics and biology*, Biofiska Institute, Bilbao, Spain
- 11/2023 *Seminar*, Basque Center for Applied Mathematics, Bilbao, Spain
- 7/2023 *Information Engines at the Frontiers of Nanoscale Thermodynamics*, Telluride, CO, USA
- 6/2023 *Webinar: Mathematics, Physics & Machine Learning*, Instituto Superior Técnico, Lisbon, Portugal
- 6/2023 *Conference: Decomposing Multivariate Information in Complex Systems* Max Planck Institute for the Physics of Complex Systems, Dresden, Germany
- 9/2022 *Seminar*, Dutch Institute of Emergent Phenomena, University of Amsterdam, Netherlands
- 8/2022 *Seminar*, Earth-Life Science Institute (ELSI), Tokyo, Japan
- 6/2022 *Evolution of complexity from the statistical physics perspective*, Yerevan Physics Institute, Armenia
- 6/2022 *Yagami Statistical Physics Seminar*, Keio University, Japan
- 5/2022 *Workshop on Stochastic Thermodynamics III* (online)

4/2022 *Cross Labs Workshop: Is AI Extending the Mind?*, Cross Labs, Japan
 2/2022 *Physics Department Colloquium Series*, University of Rochester, NY, USA
 12/2021 *Universal Biology Institute Seminar Series*, University of Tokyo, Japan
 2/2021 *Workshop: Origins of Life: The Possible and the Actual*, Santa Fe Institute, NM, USA
 7/2020 *ICTP Seminar Series*, Abdus Salam International Center for Theoretical Physics, Trieste, Italy
 5/2020 *Workshop on Stochastic Thermodynamics I* (online)
 2/2020 *AI Seminar Series*, Information Sciences Institute, Los Angeles, CA, USA
 7/2019 *ISTI Seminar Series*, Los Alamos National Lab, Los Alamos, NM, USA
 6/2018 *Connectomics Lecture Series*, Universidad Diego Portales, Santiago, Chile
 4/2018 *Meeting of the Society for the Neural Control of Movement*, Santa Fe, NM, USA
 11/2017 Seoul National University, South Korea
 8/2017 *Workshop: Thermodynamics & Computation: Towards a New Synthesis*, Santa Fe Institute, NM, USA
 10/2016 *Statistical Physics, Information Processing and Biology*, Santa Fe Institute, NM, USA
 2/2016 Information Sciences Institute, Los Angeles, CA, USA

**POPULAR
SCIENCE &
OUTREACH**

Articles

A. Kolchinsky, “The cost of sending a bit across a living cell”, *Physics Magazine*, 2023. [link](#)

Public talks

4/2018 *SITE Santa Fe* (contemporary art museum), Santa Fe, NM, USA
 Public talk on “Life, entropy, and the 2nd law of thermodynamics”

TEACHING

Online courses

2018 “Fundamentals of machine learning” (w/ B.D. Tracey), Santa Fe Institute *Complexity Explorer*
 Designed and delivered a 12-part online course on basics of machine learning ([link](#))

Workshops, lectures, and tutorials

6/2019 Santa Fe Institute Complex Systems Summer School
 Two-part lecture series on machine learning
 3/2019 Santa Fe Institute. Tutorial on “Machine learning with TensorFlow”
 6/2017, Santa Fe Institute. Co-ran tutorial on Python programming and data analysis
 6/2018
 11/2017 Seoul National University, South Korea. Co-organized week-long workshop
 “Thermodynamics, evolution, and inference through the lens of information theory”
 11/2017 ACTioN/Trustee Meeting, Santa Fe Institute
 Lecture on “Machine learning: A guide for the perplexed” (w/ B. D. Tracey)
 5/2010 Instituto Gulbenkian de Ciência, Oeiras, Portugal
 Assisted with week-long “Bayesian brain” educational module

Teaching Assistant at Indiana University, Bloomington, IN

Spring 2014 “I400 Large-scale Social Phenomena” with Simon DeDeo ([link](#))
 Spring 2011 “I201 Math and logic foundations of Informatics” with Erik Wennstrom
 Fall 2010 “I485 Biologically Inspired Computing” ([link](#)) with Luis M. Rocha
 Developed and implemented a 5-part problem set for a computational lab
 Fall 2008– “I210 Information Infrastructure” (various instructors)
 Spring 2009 Ran lab to assist with Python programming course

ADVISING

PhD dissertation committees

2025 *André Gomes*, PhD, Electrical & Computer Engineering, Instituto Superior Técnico, Lisbon, Portugal
 Thesis: “Advances in Partial Information Decomposition”

Undergraduate projects supervised

2018 *Nicolas Freitas*, Santa Fe Institute Summer REU Program, Santa Fe, NM, USA

Project: “Scaling of information in biochemical systems”

2017 *Francis Cavanna*, Santa Fe Institute Summer REU Program, Santa Fe, NM, USA

Project: “Investigating the relationship between criticality and Landauer costs using the Ising model”

**ACADEMIC
SERVICE****Grant review panels**

- European Research Council, Starting and Advanced Grant Panels
- Interdisciplinary Assessment Board for Artificial Intelligence, Spanish Research Agency
- John Templeton Foundation
- Swiss National Science Foundation

Journal refereeing

General: *PNAS*, *Nature Communications*, *npj Complexity*, *Interface Focus*

Physics: *PRL*, *PRX*, *PRX Quantum*, *PRR*, *PRE*, *NJP*, *Frontiers in Physics*, *J of Physics A*, *Physica A*, *EPL*

Biology: *PLoS Comp Bio*, *Proc R Soc B*, *Phil Trans B*, *Theory in the Biosciences*, *BioSystems*

Information theory and machine learning: *NeurIPS*, *ICLR*, *IEEE TPAMI*, *Entropy*, *Kybernetika*, *Applied Sciences*

Guest editor

Entropy special issue on “Thermodynamics and Information Theory of Living Systems” [link](#)

Conference organization

12/2025 Kyoto Workshop on Quantum Thermodynamics and Stochastic Thermodynamics, co-organizer
Kyoto University, Japan ([program](#))

10/2024 Theoretical and Experimental Approaches to Goal-Directed Behavior, co-organizer
Basque Center for Applied Mathematics, Bilbao, Spain ([press](#))

7/2024 Information-Driven States of Matter, co-organizer
University of Rochester, NY, USA ([program](#)) [press](#)

Seminar organization

2023– Co-organized ongoing “Complex Systems Lab” seminar series

2025 [Barcelona Collaboratorium for Modelling and Predictive Biology](#), Barcelona, Spain

2008– Organized a weekly discussion group on complexity and cognitive science ([link](#))

2013 Indiana University, Bloomington, IN, USA

LANGUAGES

Fluent: English, Spanish, Russian.